

AI · PHARMA COMPLIANCE

intelliCAPA

CAPA Recommendation System

Semantic search · FAISS retrieval · Explainable AI

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The Problem

CAPA management in pharmaceutical manufacturing is broken

Slow Lookup

Manual search through incident logs takes hours for rare incident types

No Semantic Search

Keyword search misses incidents described with different terminology

Inconsistency

Different engineers recommend different actions for identical incidents

No Explainability

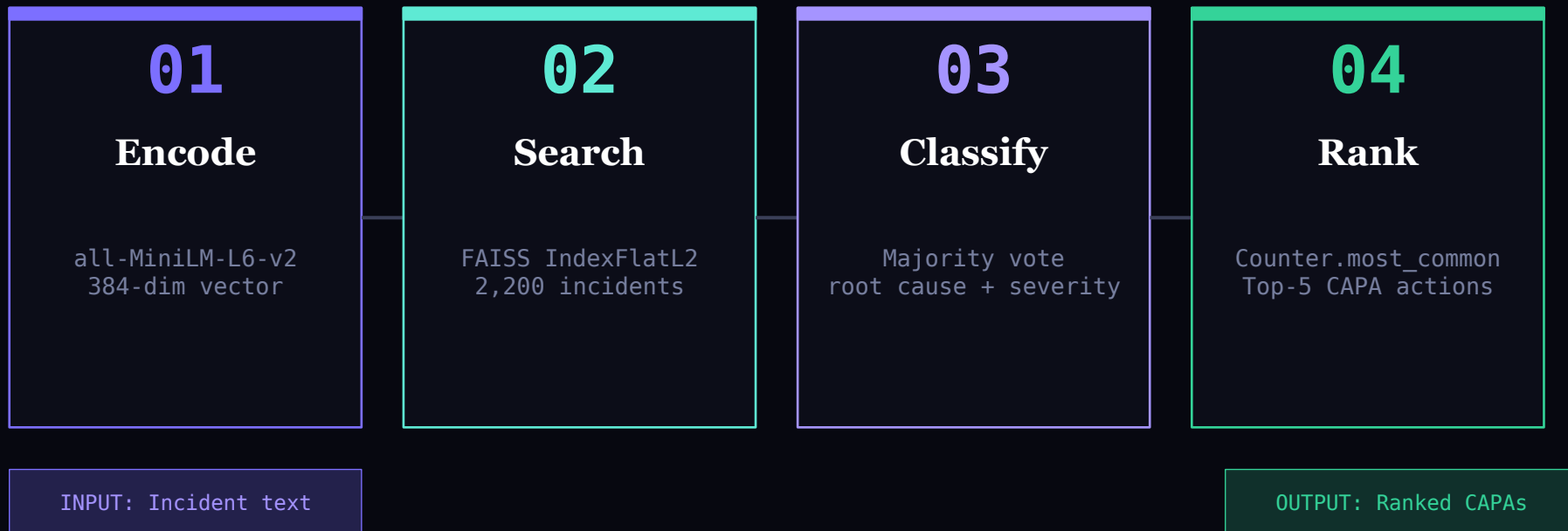
Recommendations lack traceable evidence trails for regulatory audit

Knowledge Silos

Institutional knowledge is lost when experienced staff leave the organisation

The Solution

Retrieval-based AI — no training, no GPU, sub-second response



System Architecture

CLIENT LAYER

Browser · Single-page HTML/CSS/JS app · Dark/Light mode · Chart.js analytics

API LAYER

FastAPI + uvicorn · POST /recommend · GET /health · Pydantic validation · CORS middleware

ML LAYER

SentenceTransformer all-MiniLM-L6-v2 · FAISS IndexFlatL2 · Majority voting classifier

DATA LAYER

cleaned_dataset.csv (2,200 incidents) · historical_capa_records.csv (1,000 CAPAs) · capa_index.faiss

Python 3.10+

sentence-transformers

faiss-cpu

FastAPI

uvicorn

Chart.js 4.4.1

Vanilla JS

Technical Deep Dive

● ● ● recommendation.py

```
# 1. Encode query
q = model.encode([incident])
q = np.array(q).astype('float32')

# 2. FAISS search (sub-ms)
D, I = index.search(q, k=5)

# 3. Majority vote
rc = Counter(roots).most_common(1)
sv = Counter(sevs).most_common(1)

# 4. Rank CAPA actions
corr = Counter(recs['ca']).most_common(5)
```

Embedding Module

all-MiniLM-L6-v2 loaded once at startup. encode() → 384-dim float32. Same model for index build and inference.

FAISS Retrieval

IndexFlatL2 over $2,200 \times 384 = 844,800$ floats.
index.search(q, k=5) returns L2 distances in sub-millisecond time.

Confidence Score

similarity = $\max(0, (1 - \text{dist}/2) \times 100)$. Mean across top-5 neighbours returned as recommendation confidence.

```
POST /recommend { incident_description: '...' } → 200 OK + root_cause · severity · confidence ·
corrective_actions · preventive_actions · similar_incidents
```

Evaluation Results

Leave-one-out methodology · 300 random samples · random_state=42

100%

**Root Cause
Accuracy**

All 300 test samples
correctly classified

1.000

**Precision
@ 5**

All 5 retrieved incidents
matched root cause

1.000

**MRR
@ 5**

Correct match always ranked
#1 in retrieval

95.7%

**Avg Top-1
Similarity**

Top retrieved incident near-
identical to query

⚠ **Severity Accuracy: 40%**

Expected behaviour — severity is an administrative label assigned after the incident is written. It is not encoded in the incident text and therefore not captured by semantic embedding. This is by design, not a model defect.

Features & Live System

Semantic Search

'temperature excursion' matches 'cold chain failure' — terminology doesn't matter

Analytics Dashboard

9 real-data Chart.js charts — root cause distribution, severity trends, recurrence analysis

Evidence Trail

Every recommendation traceable to specific historical records for regulatory audit

Confidence Score

Mean semantic similarity across top-5 retrieved incidents, computed per request

Add Record Modal

Extend the dataset in-app without restarting the backend

🕒 Dark / Light Mode

CSS custom properties, persisted via localStorage across sessions

Live at: intellicapa.onrender.com

Future Scope

Near Term

- PostgreSQL persistence — live DB replaces CSV so Add Record survives restarts
- PDF report export via ReportLab for formal CAPA documentation

Mid Term

- Role-based access control (QA Engineer / Manager / Admin) for 21 CFR Part 11
- Distance-weighted voting to replace majority vote with proportional influence

Long Term

- SHAP explainability — feature-level explanation of root cause classification
- Hybrid severity filtering + batch CSV upload for post-audit scenarios

Architecture is data-agnostic — identical pipeline on real production QMS records

intelliCAPA

Sub-second · Semantic · Explainable · Production-ready

100%

Root Cause Acc.

1.000

Precision@5

95.7%

Top-1 Similarity

< 1s

Response Time

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